High-level Summary Report on Preliminary ACE 2025 Data

Performance Review Unit

2026-12-16

Table of contents

Welcome **4**
IMPORTANT NOTICE 4

1 Introduction **5**

2 High-level revenues, costs, staff and balance sheet data **7**
2.1 Revenues and traffic 7
2.2 Costs 9
2.3 Staff 11
2.4 Balance sheet structure and capital expenditures 13

3 Economic cost-effectiveness **15**

4 ATM/CNS provision cost-effectiveness **17**
4.1 Pan-European system level 17
4.2 ANSP level 18

Disclaimer **22**

List of Figures

1.1 Timeline to produce the next ACE benchmarking report . . . 5
1.2 Status of 2025 data validation process . . . 6

2.1 Breakdown of gate-to-gate ANS revenues, 2025 . . . 7
2.2 Composite flight-hours of the Pan-European system, 2015–2025 . . . 8
2.3 Changes in composite flight-hours between 2015 and 2025 . . . 8
2.4 Gate-to-gate ATM/CNS provision costs at Pan-European system level, 2025 . . . 9
2.5 Changes in gate-to-gate ATM/CNS provision cost categories (real terms) . . . 10
2.6 Changes in ATM/CNS provision costs (real terms) . . . 11
2.7 Breakdown of total gate-to-gate ATM/CNS staff at Pan-European system level, 2025 . . . 12
2.8 Changes in total gate-to-gate ATM/CNS staff per staff category . . . 12
2.9 Total ANS capital & reserves and borrowings of the Pan-European system, 2015 and 2025 (real terms) . . . 13
2.10 Gate-to-gate capital expenditures of the Pan-European system, 2015 and 2025 (real terms) 14

3.1 Trend of unit economic costs at Pan-European system level, 2015-2025 (real terms) . . . 15
3.2 Economic gate-to-gate cost-effectiveness, 2025 . . . 16

4.1 Changes in unit ATM/CNS provision costs, 2015 – 2025 (real terms) . . . 17
4.2 ACE performance framework, 2025 (real terms) . . . 18
4.3 Breakdown of changes in unit ATM/CNS provision costs, 2024 – 2025 (real terms) . . . 18
4.4 ATM/CNS provision gate-to-gate cost-effectiveness, 2025 . . . 19
4.5 ATCO-hour productivity, 2025 . . . 19
4.6 Employment costs per ATCO-hour, 2025 . . . 20
4.7 Breakdown of support costs per composite flight-hour, 2025 . . . 21

Welcome

This document provides a first insight on the level of 2025 cost-effectiveness performance both for the Pan-European system and for individual ANSPs before the official release of the next ACE benchmarking report.

IMPORTANT NOTICE

Data contained in this document are preliminary and subject to changes before the publication of the final ACE benchmarking report in May 2027.

1 Introduction

The ACE benchmarking work is commissioned by the Performance Review Commission (PRC) and carried out by the EUROCONTROL Performance Review Unit (PRU) using information provided by Air Navigation Services Providers (ANSPs) in compliance with Decision No. 88 of the Permanent Commission of EUROCONTROL on economic information disclosure[1].

The data processing, analysis and reporting are conducted with the assistance of the ACE Working Group, which comprises representatives from participating ANSPs, airspace users, regulatory authorities and the Performance Review Unit. This enables participants to share experiences and establish a common understanding of underlying assumptions and data limitations.

The objective of this document is to provide a first insight on the level of 2025 cost-effectiveness performance both for the Pan-European system and for individual ANSPs before the release of the final ACE benchmarking report, which is planned end of May 2027.

Economic information disclosure by ANSPs takes time as it depends on the publication of their financial statements, which can be a lengthy process. Indeed, after the finalisation of their financial accounts, ANSPs have to follow official procedures which include, inter alia, the preparation of an independent audit report and the formal approval for publication which is granted by the relevant authorities (sometimes at Ministry level). This explains the lag between the review of the 2025 operational performance presented in the Performance Review Report (published in March 2026) and the publication of this document in December 2026.

The final ACE benchmarking report will provide more detailed information on observed changes for selected performance indicators both at Pan-European system level and at ANSP level. This detailed analysis will particularly focus on ANSPs for which significant differences.

Figure 1.1 illustrates the timeline to produce the next ACE benchmarking report.

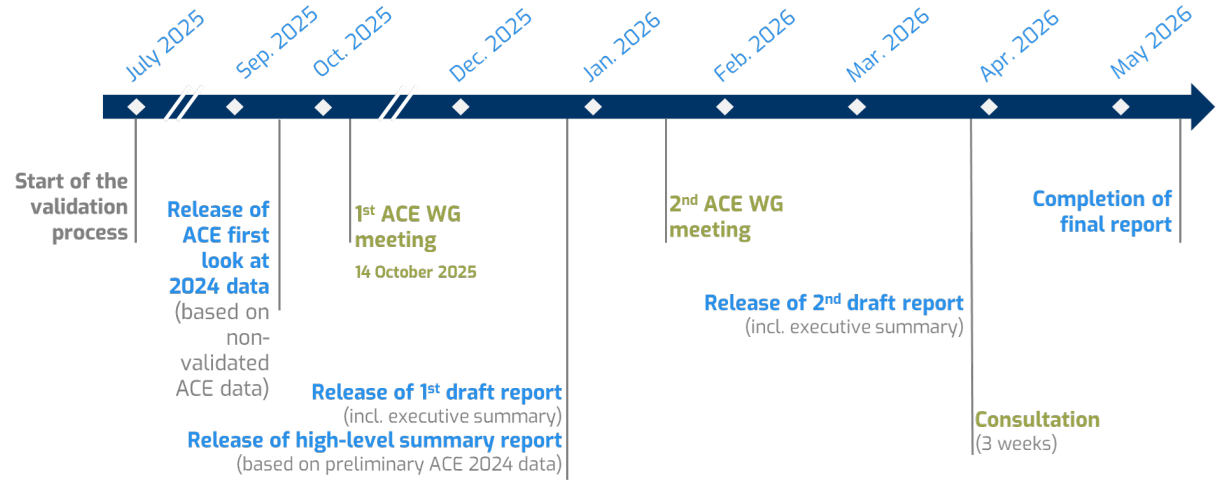


Figure 1.1: Timeline to produce the next ACE benchmarking report

It is important that robust ACE benchmarking analysis is available in a timely manner since several stakeholders, most notably ANSPs' management, regulatory authorities (e.g. NSAs) and airspace users, have a keen interest in receiving the information in the ACE reports as early as possible.

Twenty out of 38 ANSPs submitted their ACE 2025 data on time by the 1st of July 2026 and all data submissions were received by the end of September 2026. The timescale to produce the ACE benchmarking report is inevitably delayed if data are not submitted on time.

[1] Due to the on-going war in Ukraine, UkSATSE has been excluded from the ACE analysis.

It should be noted that the data presented in this document are still **preliminary and not yet fully validated**. Indeed, the data submission milestone is just the first step of a process which comprises a thorough verification and analysis of individual ANSP submissions. This validation exercise also includes a formal round of exchange between the PRU and each ANSP to ensure a common understanding of the data submitted by the ANSP.

The data used in this document reflects the information received by the 11th of November 2025. Figure 1.2 shows the status of the ACE data validation process for the data presented in this document.

Airnav Ireland ✓ (Ireland)	Croatia Control (Croatia)	Fintraffic ANS ✓ (Finland)	MATS (Malta)	ROMATSA (Romania)
Albcontrol (Albania)	DCAC Cyprus ✓ (Cyprus)	HASP (Greece)	MOLDATSA ✓ (Moldova)	Sakaeronavigatsia ✓ (Georgia)
ANS CR ✓ (Czech Republic)	DFS ✓ (Germany)	HungaroControl (Hungary)	MUAC (EUROCONTROL)	skeyes ✓ (Belgium)
ARMATS ✓ (Armenia)	DHMI ✓ (Türkiye)	LFV ✓ (Sweden)	NATS (Continental) ✓ (United Kingdom)	Skyguide (Switzerland)
Austro Control (Austria)	DSNA (France)	LGS ✓ (Latvia)	NAV Portugal (Continental) (Portugal)	Slovenia Control (Slovenia)
Avinor ANS (Continental) (Norway)	EANS ✓ (Estonia)	LPS SR (Slovakia)	NAVIAIR ✓ (Denmark)	SMATSA (Serbia/Montenegro)
BHANSA ✓ (Bosnia and Herzegovina)	ENAIRE ✓ (Spain)	LVNL ✓ (Netherlands)	Oro Navigacija (Lithuania)	
BULATSA (Bulgaria)	ENAV ✓ (Italy)	M-NAV (North Macedonia)	PANSA (Poland)	

✓ Data submission has been reviewed

Figure 1.2: Status of 2025 data validation process

The data contained in this report is therefore subject to changes before the release of the final ACE benchmarking report in May 2027.

The remainder of this report is structured as follows:

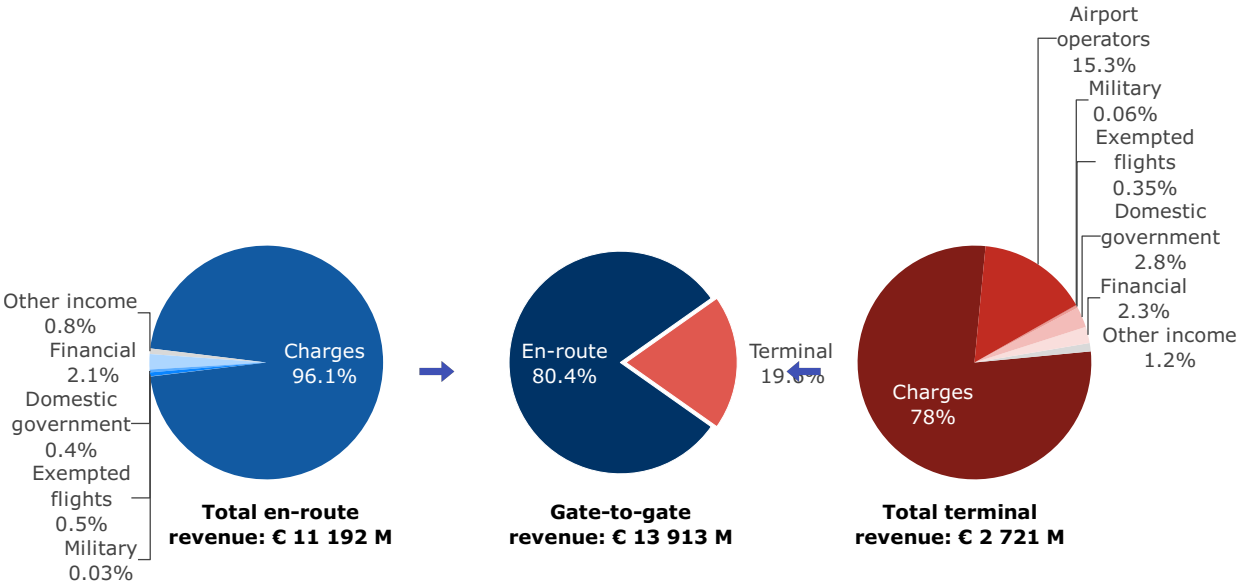
- Chapter 2: provides a high-level presentation of 2025 revenues, costs, staff and balance sheet data.
- Chapter 3: presents a preliminary analysis of economic cost-effectiveness at Pan-European and ANSP level.
- Chapter 4: presents a preliminary analysis of ATM/CNS provision cost-effectiveness at Pan-European and ANSP level, and underlying components.

2 High-level revenues, costs, staff and balance sheet data

2.1 Revenues and traffic

This chapter provides a preliminary overview of high-level revenues, costs and staff data provided in ANSPs ACE 2025 data submissions.

Total ANS revenues in 2025 amounted to €13 913M. Most en-route revenues come from the collection of en-route charges (96.1%, see left pie chart). The proportion of terminal revenues from charges is lower (78.0%, see right pie chart), as additional income may directly come from airport operators (15.3%) through, for example, a contractual arrangement between the ANSP and the airport operator.



En-route	%	Gate-togate revenues (€ M)	%	Terminal
10 761	96.1%	Income from charges	78%	2 124
n.a.	n.a.	Income from airport operators	15.3%	416
4	0.03%	Income from the military	0.06%	2
60	0.5%	Income in respect of exempted flights	0.35%	9
44	0.4%	Income from domestic goverment	2.8%	77
237	2.1%	Financial income	2.3%	62
86	0.8%	Other income (incl. exceptional revenue item)	1.2%	31
11 192	100%		100%	2 721

Figure 2.1: Breakdown of gate-to-gate ANS revenues, 2025

Across the Pan-European system, traffic in 2025 (measured in composite flight-hours) increased by +4.9% compared to 2024.

Taking a longer-term perspective and setting aside the exceptional disruption to air transport during the COVID-19 pandemic (mainly in 2020 and 2021), at Pan-European system level traffic continued its upward trend, with the level recorded in 2025 being +22.5% higher than in 2015 and +6.5% higher than in 2019.

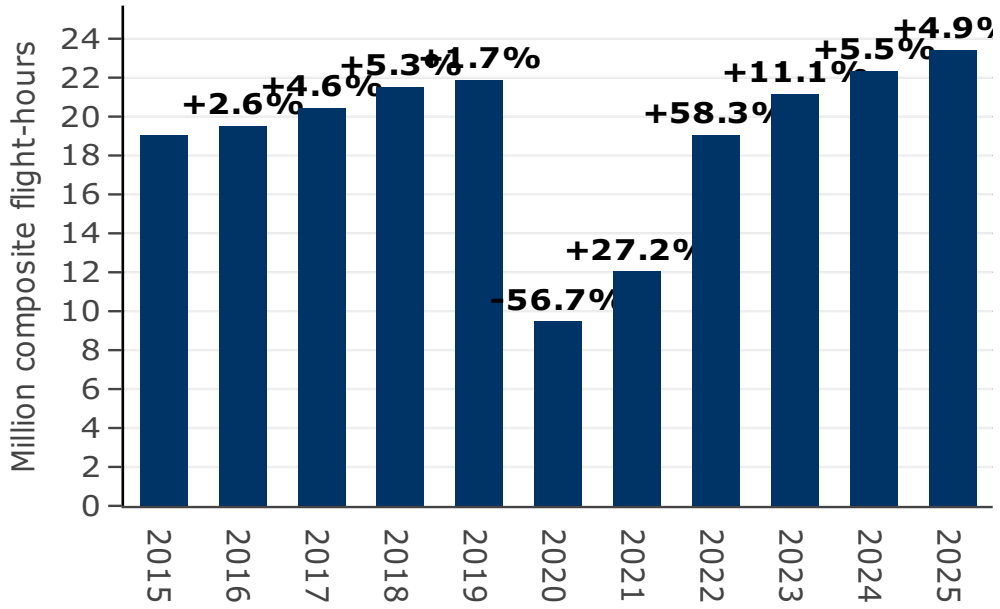


Figure 2.2: Composite flight-hours of the Pan-European system, 2015–2025

However, at ANSP level, traffic flows and characteristics changed significantly over the 2015-2025 period, influenced by several factors. These included a) the ongoing war in Ukraine, which led to significant changes in route structures and traffic volumes due to airspace closures and reciprocal sanctions on air carriers; and b) other developments reflecting evolving demand for air transport and changes in how airspace users provide their services (e.g. increasing demand for holiday destinations, more pronounced seasonality, use of heavier aircraft flying longer routes, fewer domestic flights in some countries, etc.)[1].

As a result, eight ANSPs were still below their 2015 traffic levels in 2025 (see Figure 2.3) and 16 below 2019 levels. This inevitably impacted the levels and trends of ACE indicators for the most affected ANSPs.

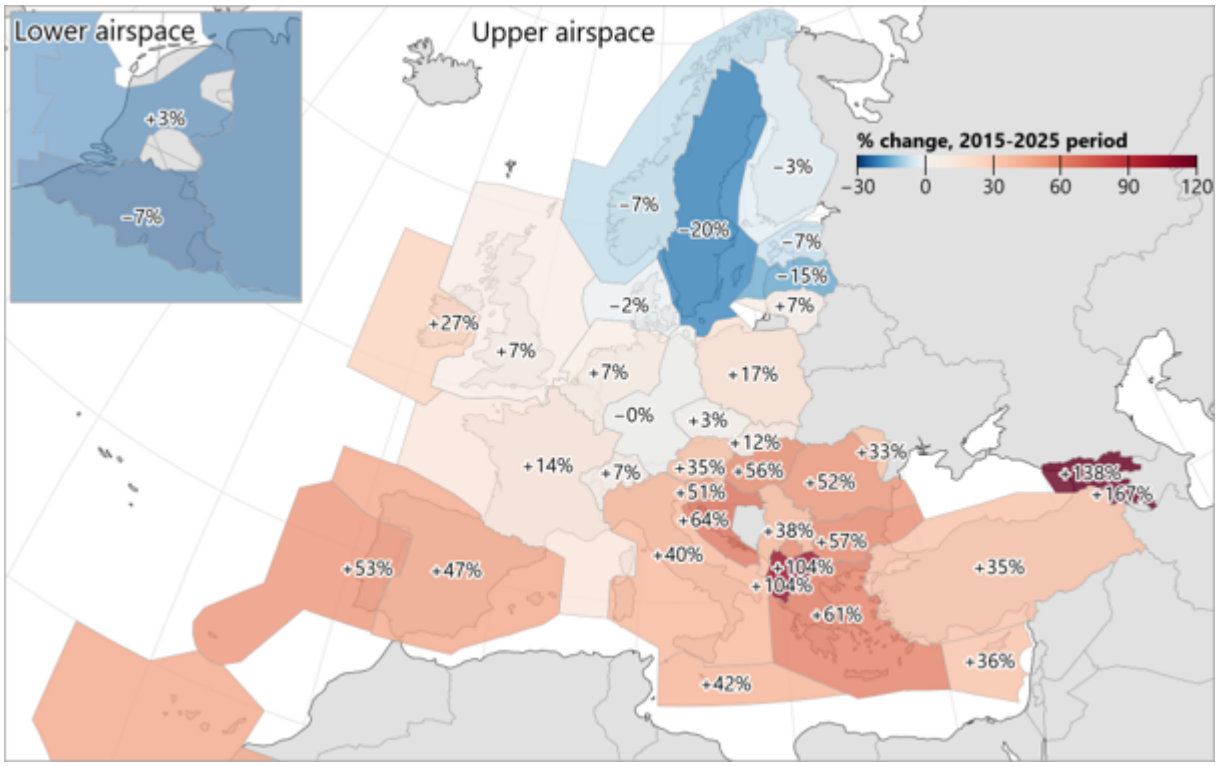
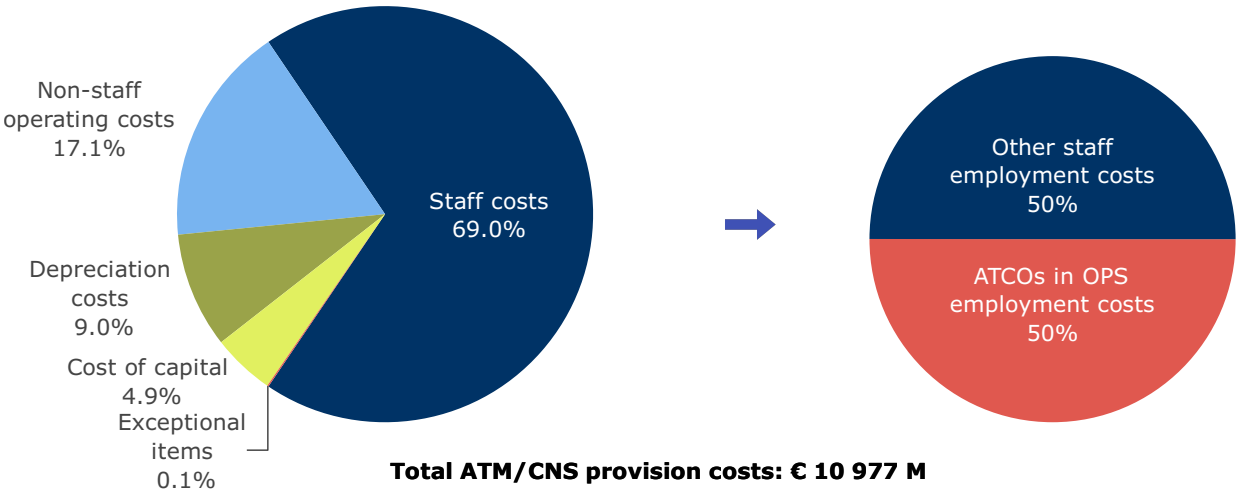


Figure 2.3: Changes in composite flight-hours between 2015 and 2025

[1] Performance Review Report, March 2026.

2.2 Costs

The ACE benchmarking analysis focuses on the specific costs of providing gate-to-gate ATM/CNS services, which amounted to €10 468M in 2025. Operating costs (including staff costs, non-staff operating costs and exceptional cost items) accounted for some 84% of total ATM/CNS provision costs, while depreciation costs and the cost of capital represented around 16%.



	En-route		Terminal		Gate-to-gate	
	€ M	%	€ M	%	€ M	%
Staff costs	5 880	68.3%	1 689	71.3%	7 569	69.0%
ATCOs in OPS employment costs	2 876	n.a.	910	n.a.	3 786	n.a.
Other staff employment costs	3 004	n.a.	780	n.a.	3 783	n.a.
Non-staff operating costs	1 454	16.9%	423	17.9%	1 877	17.1%
Depreciation costs	811	9.4%	175	7.4%	986	9.0%
Cost of capital	451	5.2%	83	3.5%	534	4.9%
Exceptional items	11	0.1%	0	0.0%	11	0.1%
Total ATM/CNS provision costs	8 607	100.0%	2 370	100.0%	10 977	100.0%

Figure 2.4: Gate-to-gate ATM/CNS provision costs at Pan-European system level, 2025

Between 2024 and 2025, ATM/CNS provision costs increased by +2.0% (+€210M). The main drivers were higher staff costs (+€417M) while all other cost categories decreased.

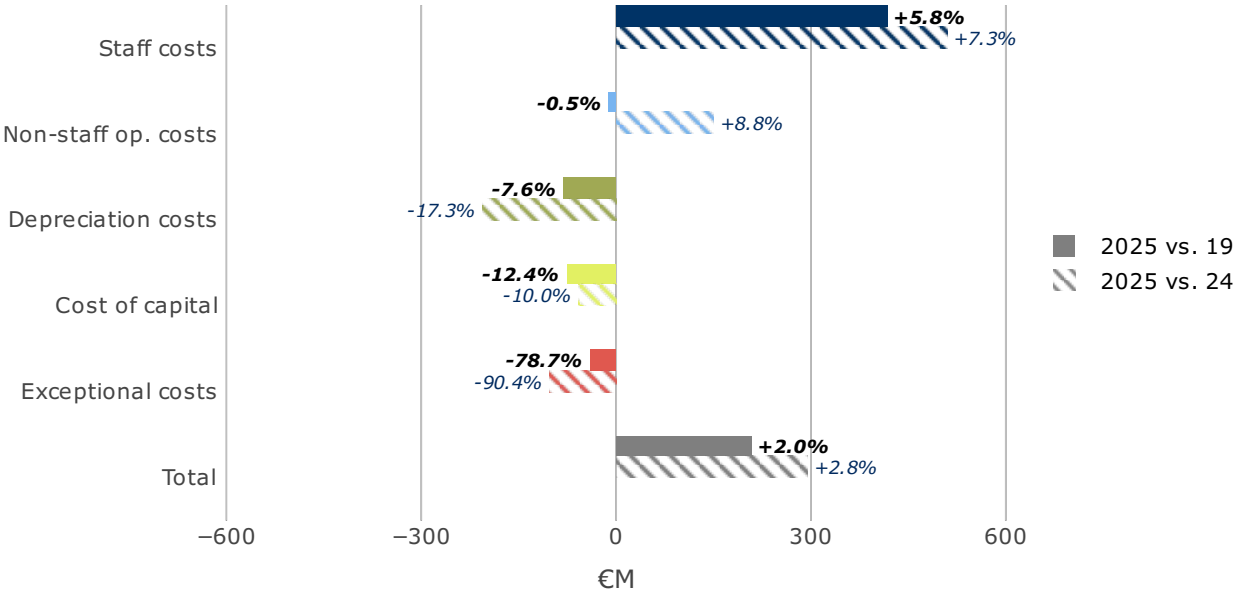
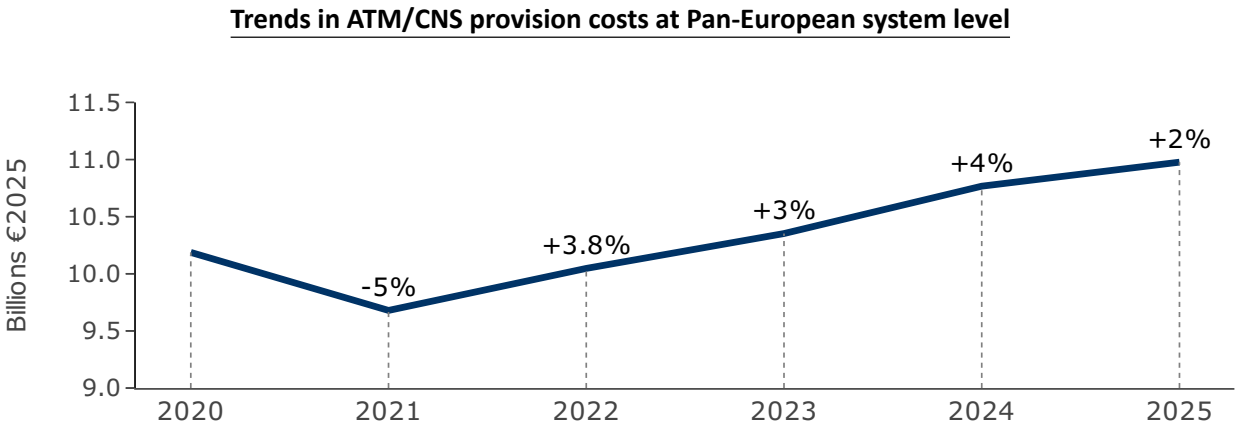


Figure 2.5: Changes in gate-to-gate ATM/CNS provision cost categories (real terms)

In 2025, the six largest ANSPs in terms of costs (DSNA, DFS, ENAIRE, NATS, ENAV and DHMI) bore some 57% of the total Pan-European gate-to-gate ATM/CNS provision costs, while the six smallest ANSPs accounted for some 1% (see bottom left part of Figure 2.6).



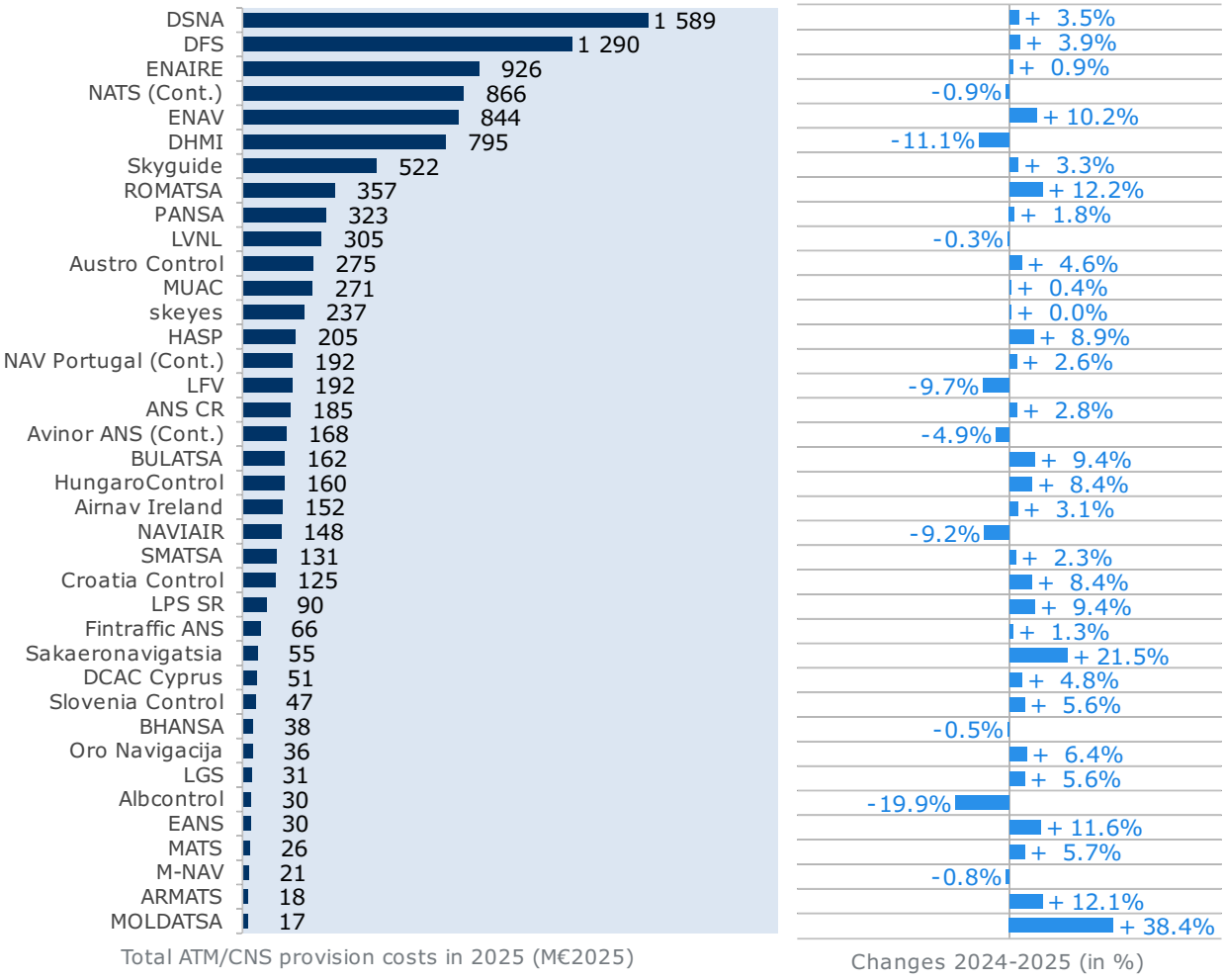


Figure 2.6: Changes in ATM/CNS provision costs (real terms)

2.3 Staff

The Pan-European ANSPs employed a total of 56 146 staff (expressed in Full Time Equivalents – FTEs) in 2025 (comprising 55 309 staff providing ATM/CNS services and 837 internal MET staff). Some 17 840 staff (32%) were ATCOs working on operational duties, split between ACCs (53%) and APP/TWR facilities (47%). On average, 2.1 additional staff were required for every ATCO in OPS in Europe.

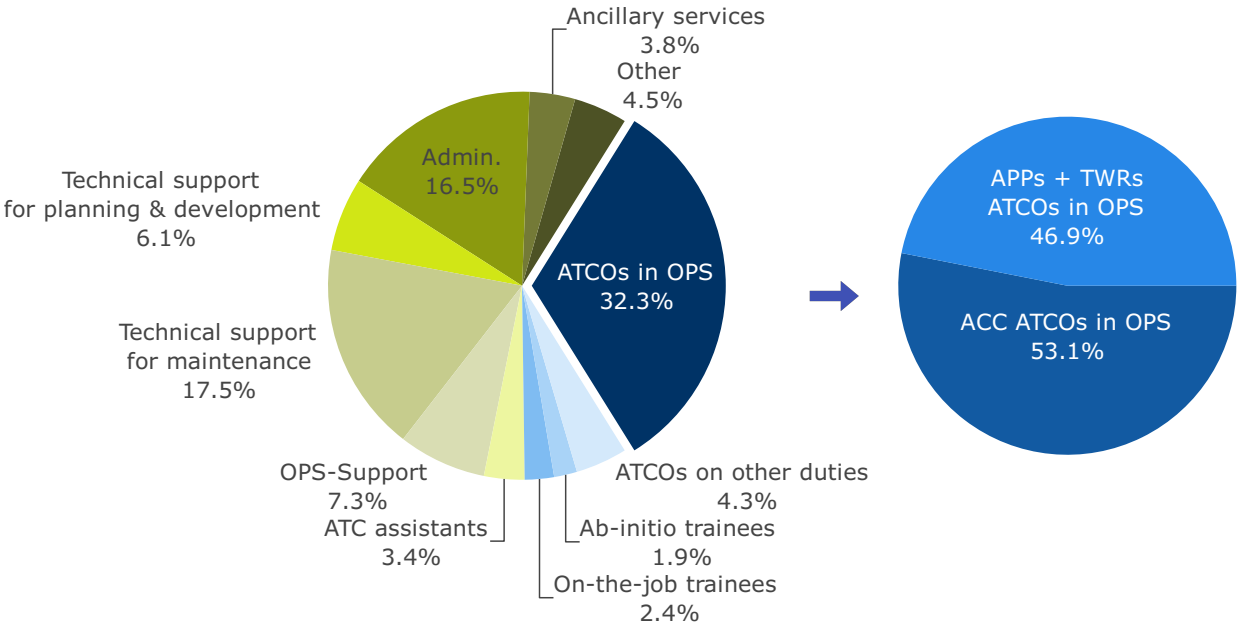


Figure 2.7: Breakdown of total gate-to-gate ATM/CNS staff at Pan-European system level, 2025

In 2025, the number of ATM/CNS staff increased by +2.4% (+1 278 FTEs) compared with 2024, reaching a level +8.0% above that recorded in 2015. The numbers of ATCOs in OPS and support staff rose by +1.5% (+264 FTEs) and +2.8% (+1 014 FTEs), respectively, compared with 2024. The increase in ATCOs in OPS at the Pan-European system level was mainly attributable to a change in ENAV's reporting scope. In 2025, ENAV included 20 additional tower units serving regional airports with relatively low traffic which were previously excluded from the ACE benchmarking exercise. Excluding ENAV from the 2024-2025 comparison, the increase in the number of ATCOs in OPS would be lower (+0.4%).

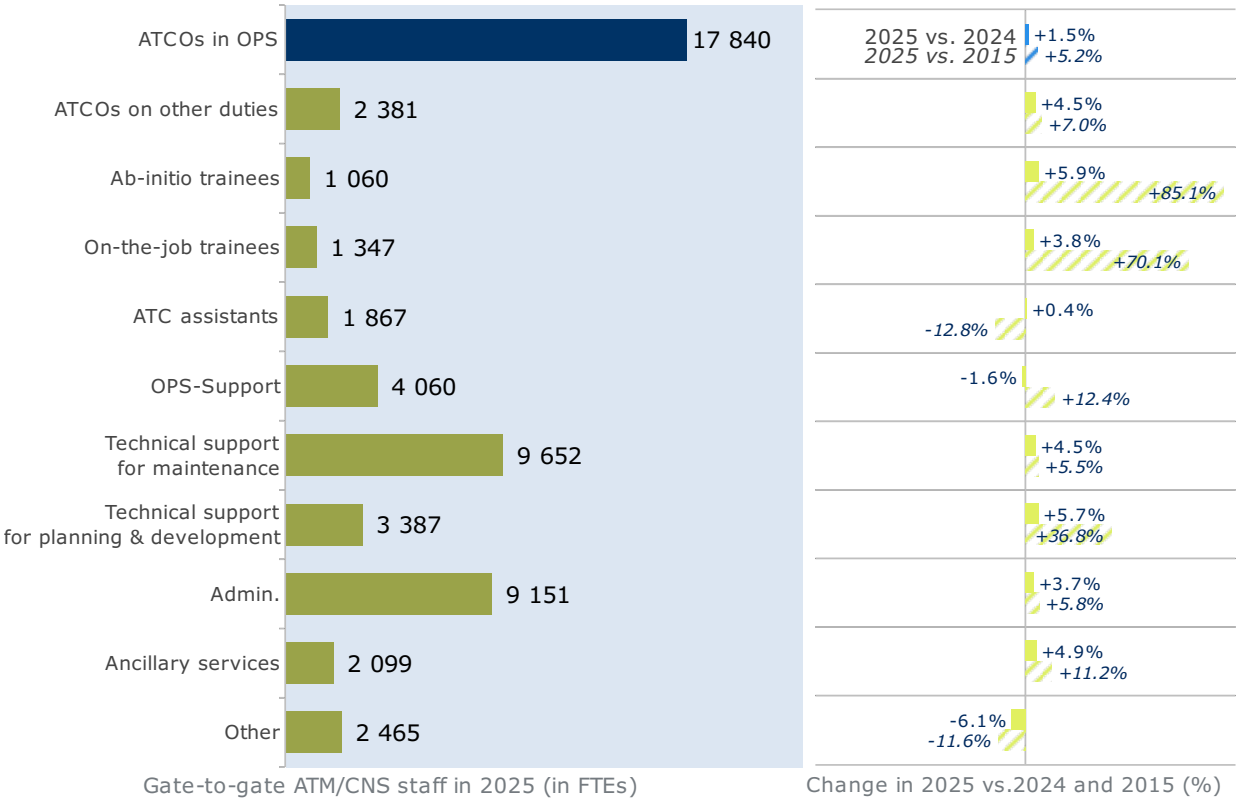


Figure 2.8: Changes in total gate-to-gate ATM/CNS staff per staff category

The overall change in staff numbers between 2024 and 2025 was mainly driven by the following categories:

- Technical support staff for operational maintenance, monitoring and control: +413 FTEs (+4.5%);
- Administrative staff: +328 FTEs (+3.7%);
- ATCOs in OPS: +264 FTEs (+1.5%); and,
- Technical support staff for planning and development: +183 FTEs (+5.7%).

Most other categories recorded increases, except OPS support (non-ATCOs) and Other Staff, which declined by -1.6% and -6.1%, respectively, as shown in Figure 2.8. The largest proportional increase was observed in ab-initio trainees (+5.9%), indicating that some ANSPs expanded the recruitment of future ATCOs in 2025. More detailed information on staff numbers will be provided in the upcoming ACE report.

2.4 Balance sheet structure and capital expenditures

ANSP balance sheets in 2025 comprised €10 017M in capital and reserves and €5 091M in borrowings, while capital expenditures amounted to €1 454M.

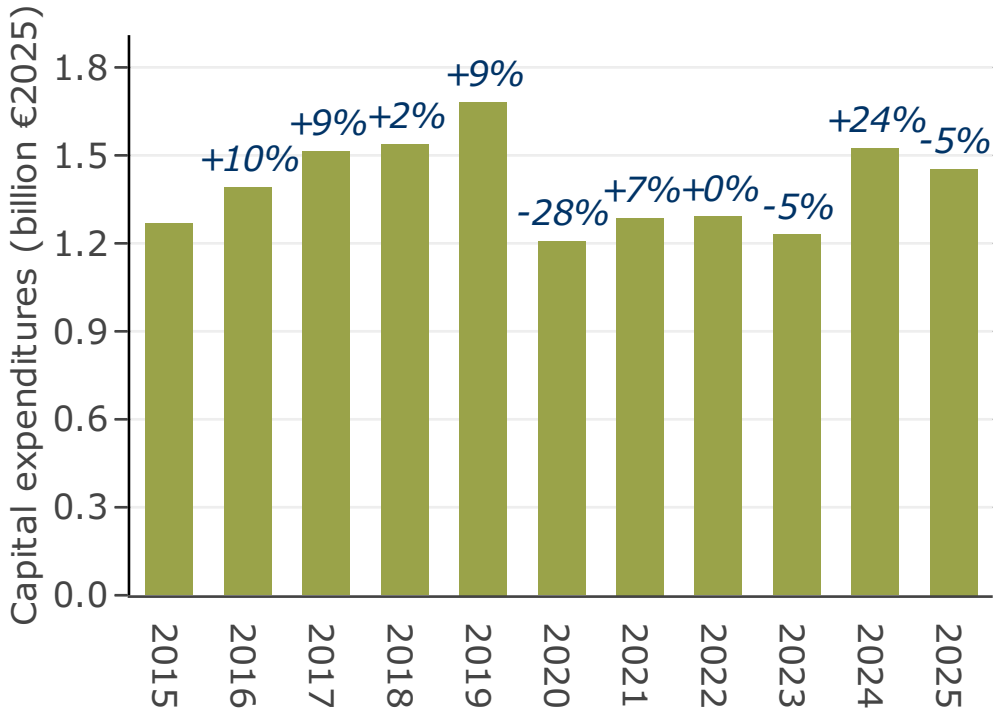


Figure 2.9: Total ANS capital & reserves and borrowings of the Pan-European system, 2015 and 2025 (real terms)

In 2025, capital and reserves decreased by -6% (€649M), while borrowings continued to decline, falling by -9% (-€483M). At €5.1 billion, borrowings remained significantly above their 2019 level (the lowest recorded over the 2015-2025 period) but continued to move closer to their 2015 level of €4.0 billion.

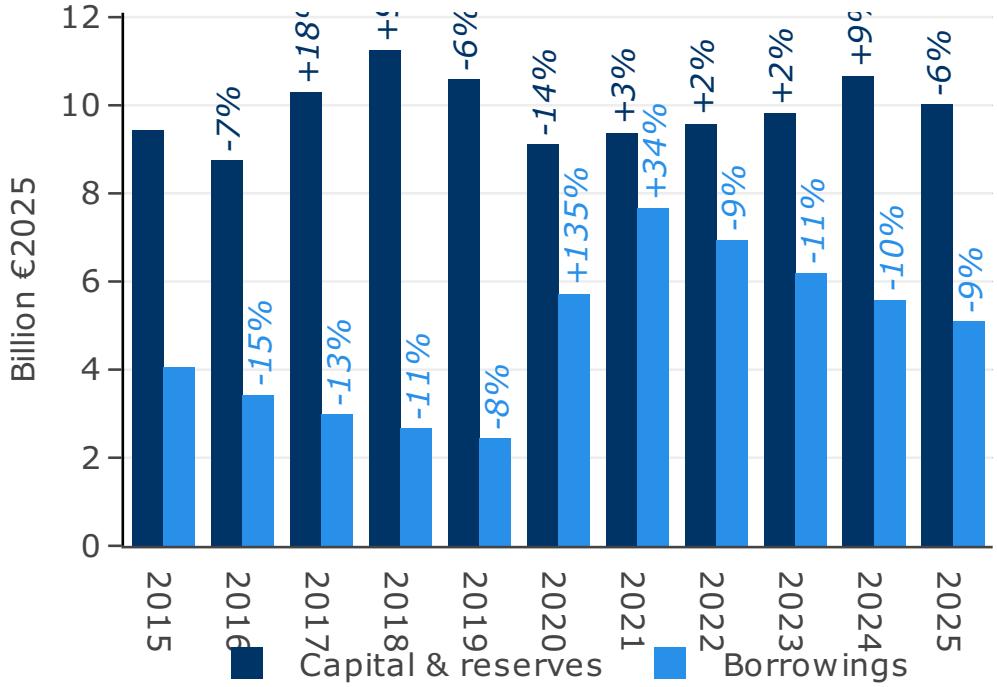


Figure 2.10: Gate-to-gate capital expenditures of the Pan-European system, 2015 and 2025 (real terms)

Although capital expenditures fell in 2025, they remained close to their 2024 level (€1.5 billion) and well above the amounts spent during 2020-2023, a period marked by liquidity constraints and considerable uncertainty about future traffic volumes.

3 Economic cost-effectiveness

The concept of economic cost-effectiveness, developed by the PRC, is defined as the sum of gate-to-gate ATM/CNS provision costs and the costs of ground ATFM delays for both en-route and airport, all expressed per composite flight-hour. This economic performance indicator is meant to capture trade-offs between quality of service provided and costs[1].

Figure 3.1 shows preliminary results on the changes in economic cost-effectiveness over 2015 - 2025 at Pan-European system level.

Unit economic costs fell by -7.1% in 2025, driven by a -18.8% decrease in the unit costs of ATFM delays, and a -2.8% reduction in ATM/CNS provision costs per composite flight-hour. Although gate-to-gate ATFM delays fell from 29.6 million minutes in 2024 to 25.2 million minutes in 2025, they remained relatively high, with estimated costs of €3.4 billion. ATFM delay costs accounted for 23% of unit economic costs, down from the record high of 27% in 2024 but close to the shares recorded in 2018, 2019 and 2023.

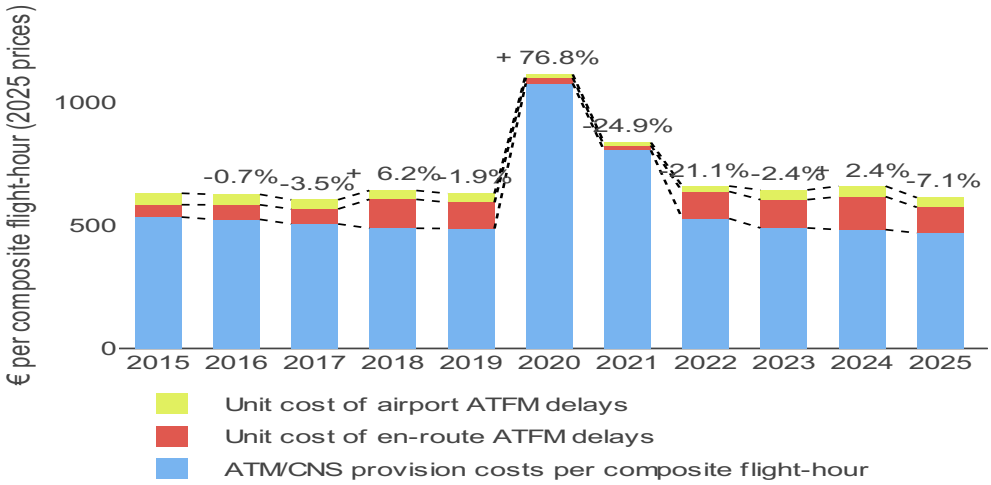


Figure 3.1: Trend of unit economic costs at Pan-European system level, 2015-2025 (real terms)

Figure 3.2 shows preliminary results at ANSP level (dotted lines represent the 1st and 3rd quartiles, €433 and €928, respectively).

[1] See https://ansperformance.eu/economics/ace/ace-handbook/ for more information on the methodology used to compute composite flight-hours and economic costs.

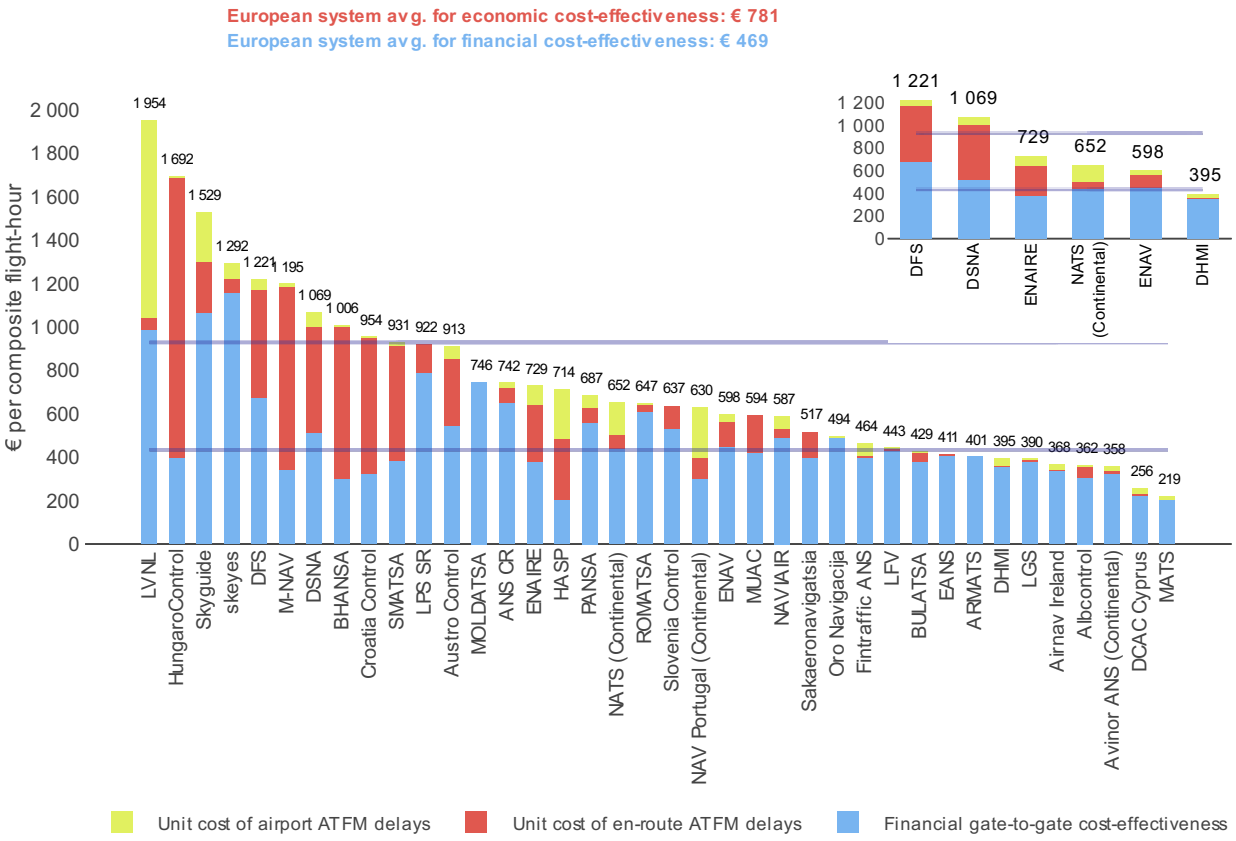


Figure 3.2: Economic gate-to-gate cost-effectiveness, 2025

For ten ANSPs ATFM delays represented more than 30% of their unit economic costs (see ANSPs with the largest red and yellow portions, e.g. M-NAV, BHANSA, HASP, SMATSA, DSNA, etc.). In absolute terms (cumulative ATFM delays in minutes) DSNA, ENAIRE and HASP generated 50% of the Pan-European system total ATFM delays.

The primary sources of ATFM delays in 2025 were capacity and staffing shortages (47% of total ATFM delays), followed by adverse weather conditions (34% of total ATFM delays).

Further analysis of the relationship between changes in ANSPs costs, traffic and unit costs will be presented in the forthcoming ACE report.

4 ATM/CNS provision cost-effectiveness

This chapter provides a preliminary analysis of ATM/CNS provision cost-effectiveness.

4.1 Pan-European system level

Figure 4.1 shows that ATM/CNS costs per composite flight-hour stood at €469 in 2025, which represents the lowest level since the inception of ACE benchmarking activities in 2002. Compared to 2024, unit ATM/CNS provision costs were -2.8% lower, as traffic (+4.9%) rose faster than ATM/CNS provision costs (+2.0%).

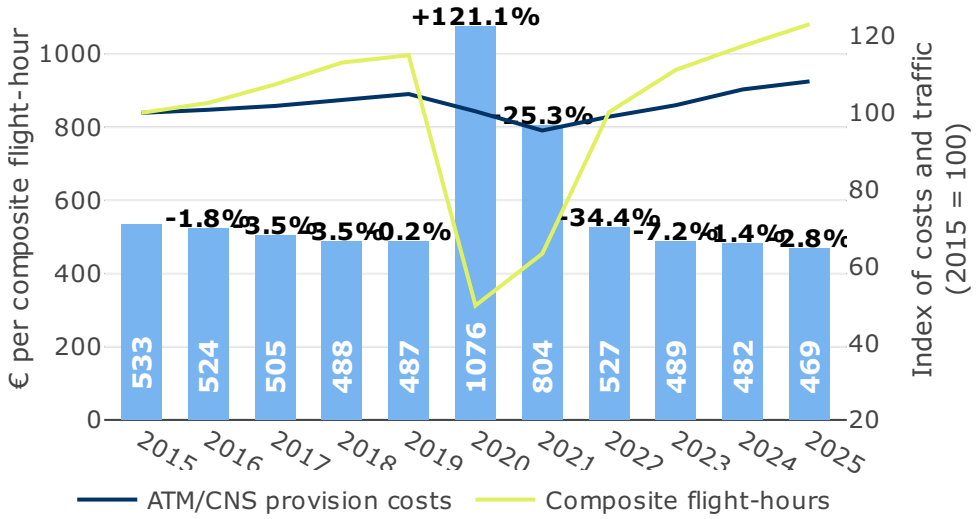


Figure 4.1: Changes in unit ATM/CNS provision costs, 2015 – 2025 (real terms)

The analytical framework used in the ACE analysis to break down the ATM/CNS provision cost-effectiveness indicator into relevant economic drivers is presented in Figure 4.2. These key drivers include:

a) ATCO-hour productivity (1.00 composite flight-hours per ATCO-hour);

b) ATCO employment costs per ATCO-hour (€162); and,

c) support costs per unit output (€307).

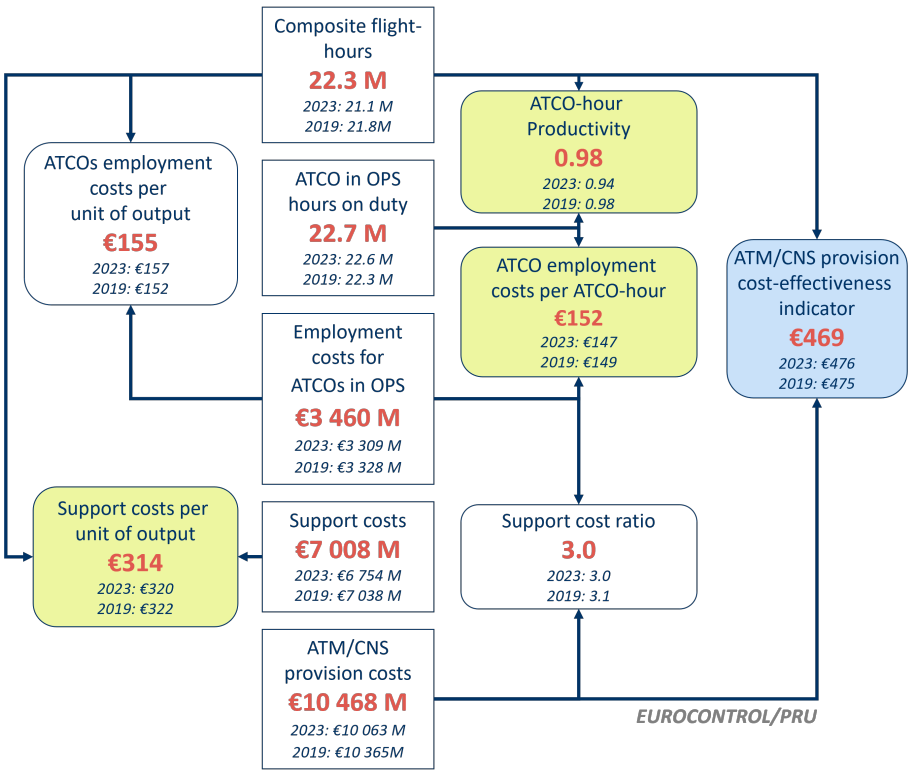


Figure 4.2: ACE performance framework, 2025 (real terms)

Figure 4.3 shows that in 2025, ATCO employment costs per ATCO-hour rose by +3.3% while ATCO-hour productivity rose by +2.2% compared to 2024. As a result, ATCO employment costs per composite flight-hour increased by +1.1%. In the meantime, unit support costs fell by 4.7% due to an increase in composite flight-hours (+4.9%) while total support costs remained fairly stable (-0.1%). Consequently, in 2025, unit ATM/CNS provision costs fell by -2.8% at the Pan-European system level compared to 2024, taking into account the relative weighting of ATCO employment and support costs based on their shares of total ATM/CNS provision costs.

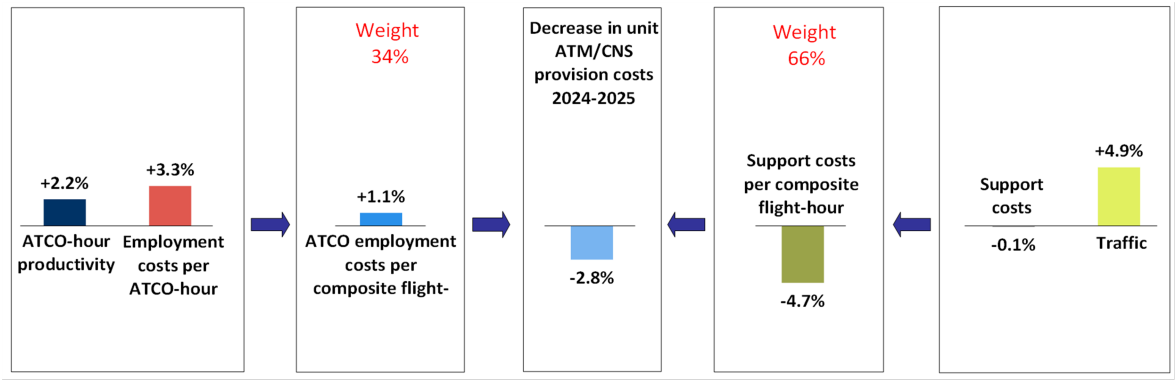


Figure 4.3: Breakdown of changes in unit ATM/CNS provision costs, 2024 – 2025 (real terms)

4.2 ANSP level

All figures presented in this section show the preliminary benchmarking results for the 38 ANSPs. Because of their weight in the Pan-European system, the six largest ANSPs by cost (DFS, DHMI, DSNA, ENAIRE, ENAV and NATS) are also shown in a miniature replica of the chart (top right corner of the figures). The 1st and 3rd quartiles for each indicator are also shown in all figures. The gap between these two quartiles provides additional insight on the dispersion of the values.

Many factors contribute to observed differences in ANSPs' performance. Over the years, the Performance Review Unit has identified factors which can be of an exogenous nature (i.e. those outside the control of an ANSP) or of an endogenous nature (those entirely under the ANSP's control). A detailed list of these factors can be found in the ACE Handbook.

Figure 4.4 presents the ATM/CNS provision gate-to-gate cost-effectiveness indicator at ANSP level for the year 2025. The dotted lines represent the 1st and 3rd quartiles (€348 and €542, respectively).

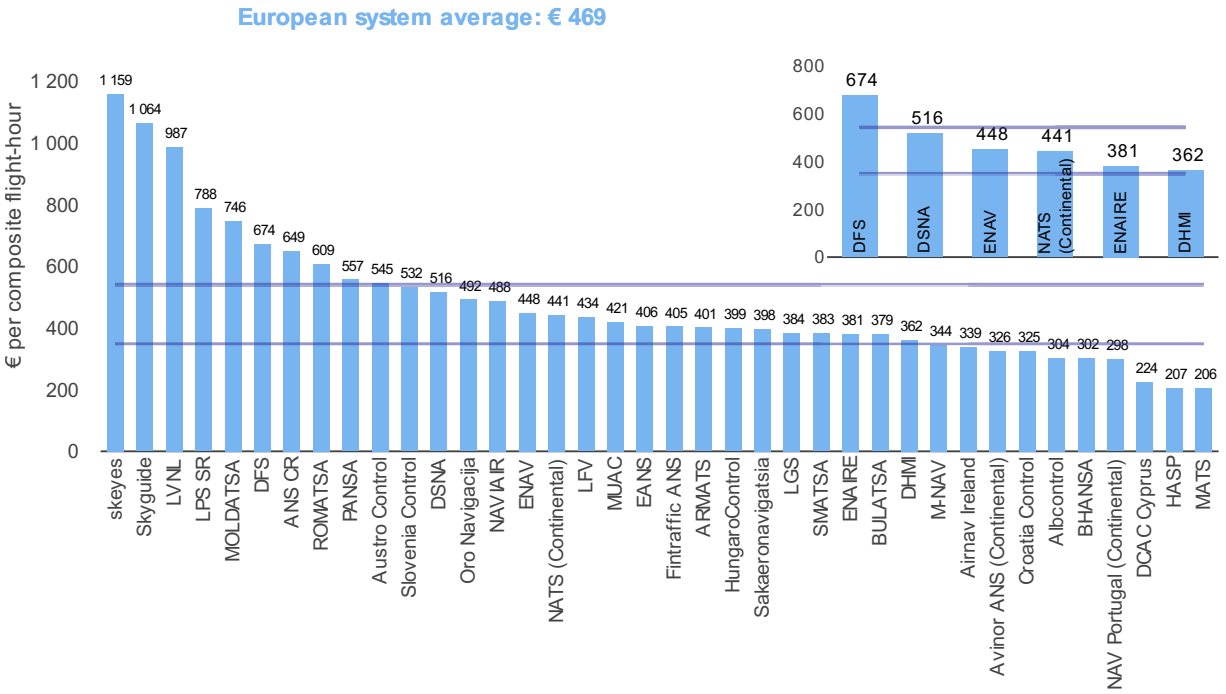


Figure 4.4: ATM/CNS provision gate-to-gate cost-effectiveness, 2025

Figure 4.5 presents the ATCO-hour productivity indicator at ANSP level for the year 2025. The dotted lines represent the 1st and 3rd quartiles (0.8 and 1.08, respectively).

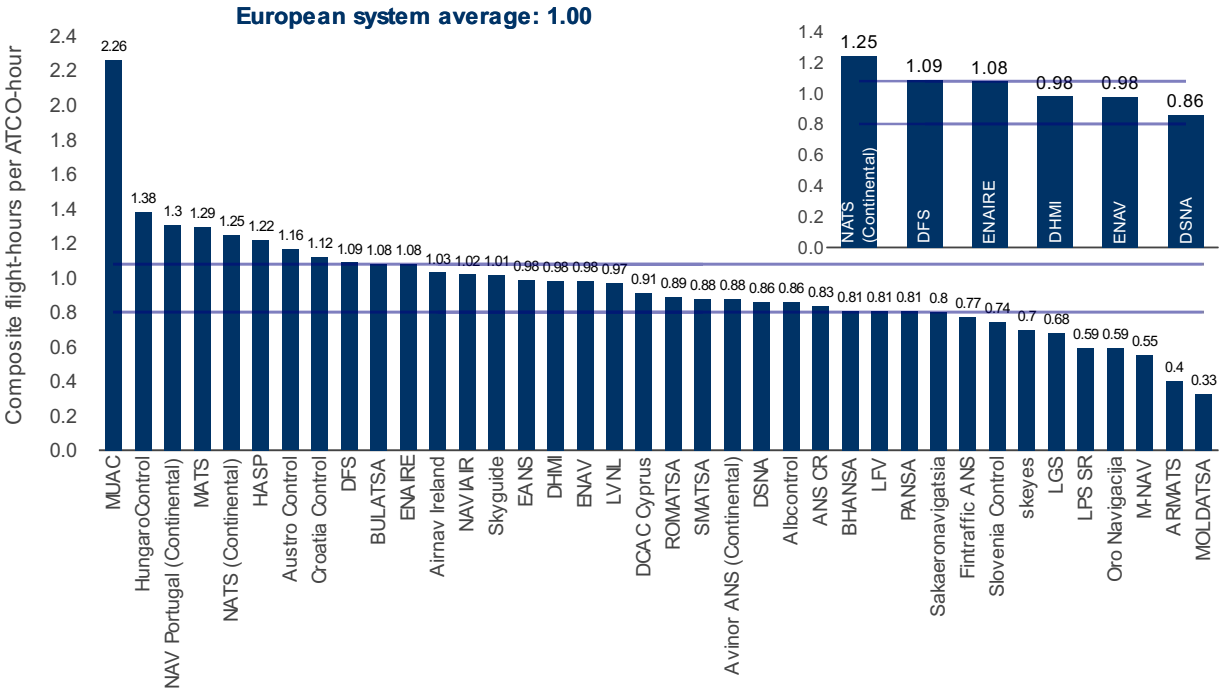


Figure 4.5: ATCO-hour productivity, 2025

ATCO-hour productivity varies significantly across the ACE sample. Observed differences should be considered together with other indicators such as ATCO employment and support costs. Multiple factors influence productivity, including operational conditions (traffic levels, airspace type, number of airports), legal and socio-economic aspects (labour laws), and institutional factors (e.g. governance arrangements). More details on the ATCO-hour productivity indicator will be provided in the ACE benchmarking report.

Figure 4.6 presents the employment costs per ATCO in OPS indicator at ANSP level for the year 2025. The dotted lines represent the 1st and 3rd quartiles (€81 and €171, respectively).

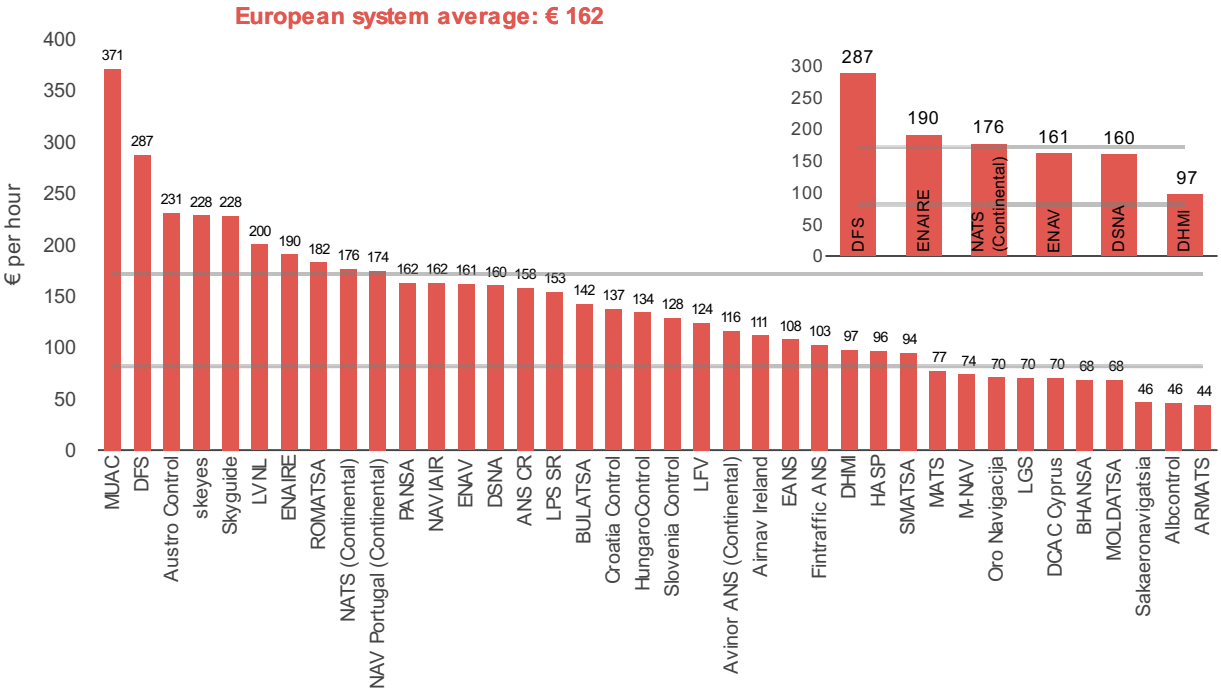


Figure 4.6: Employment costs per ATCO-hour, 2025

Figure 4.7 presents the support costs per composite flight-hour indicator at ANSP level for the year 2025. The dotted lines represent the 1st and 3rd quartiles (€235 and €358, respectively).

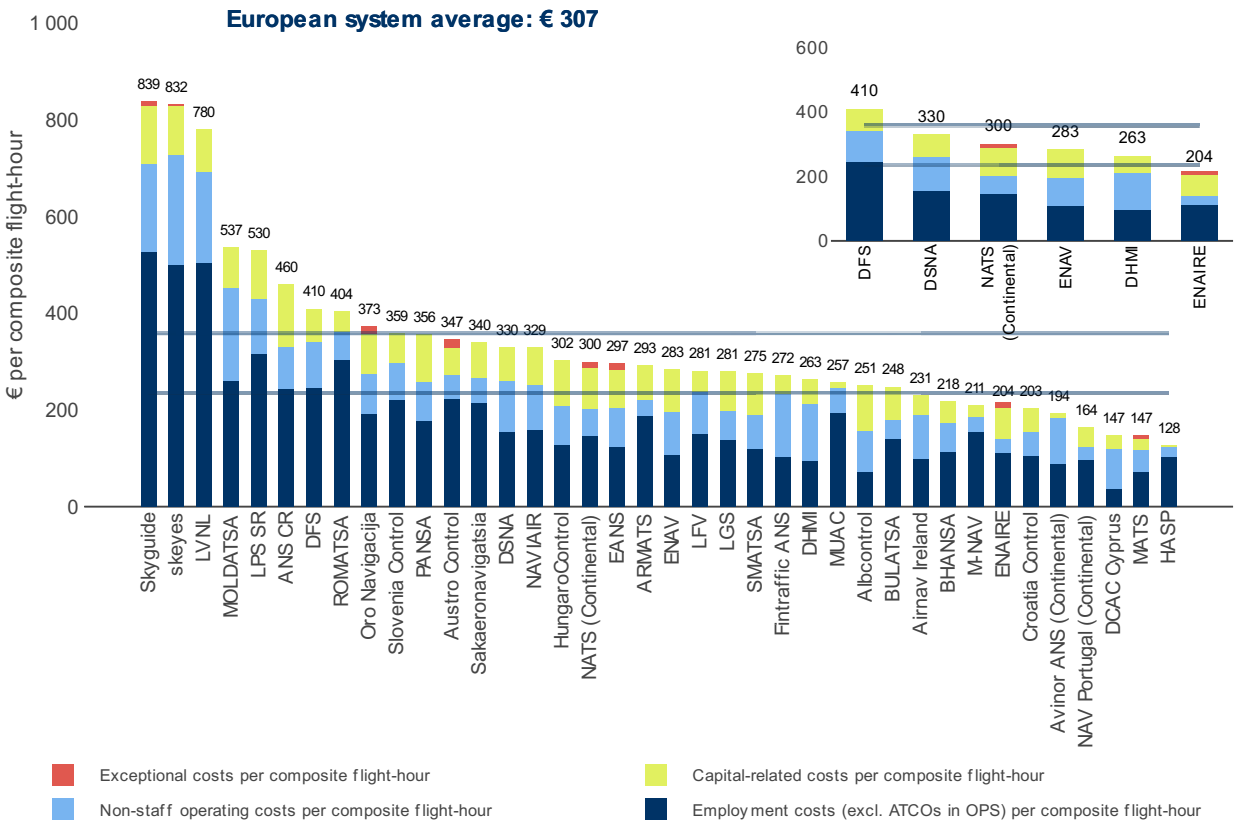


Figure 4.7: Breakdown of support costs per composite flight-hour, 2025

A more detailed analysis of the changes in cost-effectiveness, ATCO-hour productivity, ATCO employment costs per ATCO-hour and unit support costs will be available in the final ACE benchmarking report.

Disclaimer

The Performance Review Unit (PRU) has made every effort to ensure that the information and analysis contained in this document are as accurate and complete as possible. Should you find any errors or inconsistencies we would be grateful if you could please bring them to the PRU's attention. The PRU's e-mail address is pru-support@eurocontrol.int